

Homy Xu

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RESEARCH INTERESTS

My research interests lie in **coding foundation models, LLM agents, and memory systems**, with a broader background in **database systems and Text-to-SQL**. I study long-horizon agent execution, state-consistent memory, and training-data quality through system development and benchmark-driven evaluation.

EDUCATION

Shanghai Jiao Tong University

Undergraduate in Information Security, School of Computer Science

Member of the Zhiyuan Honors Program

Shanghai, China

Sept. 2024 – Present

Selected Coursework: Data Structures (Honors); Programming Concepts and Methodology (Honors, C++); Compiler Principles; Operating Systems; Computer Organization and Architecture; Programming Practice; Mathematical Analysis (Honors); Linear Algebra (Honors).

SELECTED PUBLICATIONS & MANUSCRIPTS

MemTrace: State-Consistent Memory for Long-Horizon Coding Agents

Under review at ICLR | **First author** [[Repository](#)]

Aug. 2026 – Present

Developed a state-consistent memory system that combines structured execution traces with repository grounding to validate historical evidence before reuse. Evaluated on three software-engineering benchmarks using two agent harnesses.

Dial: A Knowledge-Grounded Dialect-Specific NL2SQL System

PVLDB 2026 (accepted) | **Co-first author (listed second)** [[arXiv](#)] [[Code](#)]

Aug. 2025 – Mar. 2026

Contributed to natural language preprocessing, dialect-aware logical planning, and knowledge retrieval for multi-dialect NL2SQL. Helped design and initialize **HINT-KB**, prepare datasets, and develop iterative feedback and semantic correction; participated in failure analysis and rebuttal revisions.

Are We Ready for An Agent-Native Memory System?

Under review at VLDB [[arXiv](#)] [[Code](#)]

Mar. 2026 – June 2026

Contributed to a systematic study of agent memory across representation and storage, extraction, retrieval/routing, and maintenance. Evaluated task effectiveness, retrieval fidelity, dynamic-update robustness, long-term stability, and runtime cost across diverse workloads.

Tensor-Network Physical Pages for Quantum-Ready Data Management

QCDKM 2026 (accepted)

May 2026 – June 2026

Contributed to **TN-Page**, a tensor-network physical-page representation for approximate data management and quantum backends. Studied workload-aware bucketization, tensorization, factorization, and backend-aware encoding, with evaluation of fidelity, compression, qubit usage, and gate depth.

EXPERIENCE

MemTensor

Research Intern — Mentor: Juncheng Zhang (Tsinghua University)

Aug. 2026 – Present

- Research coding foundation models and agents, focusing on training-data quality, trajectory-based learning, and repository-level evaluation of code editing, testing, and debugging.
- Completed the first-author **MemTrace** study, developing and evaluating state-consistent memory for reliable execution of long-horizon coding tasks.

HONORS

Zhiyuan Honors Scholarship

SKILLS

Programming & Systems: C++, SQL, Git/GitHub, LaTeX; data structures and algorithms, operating systems, and computer architecture.

Research: Text-to-SQL, LLM agents, memory systems, RAG, benchmark design, experimental analysis, ablation and failure analysis, manuscript preparation, and rebuttal collaboration.